

Code Interpretation: Tombe & Zhu (2019, *AER*)

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1 initial_eqm.m

```
1 close all
2 clear all
3 global N theta beta_ag beta_na alpha eta_ag eta_na gamma kappa dni_ag ...
   dni_na pi_ag pi_na mu L La_data Ln_data Libase drYL_data La Ln
```

```
1 % Merge in the mu and price data
2 global drYL_data
3 data=importdata('employment_realGDP_data.csv'); % use data.csv for the ...
   BTZ version; "data - official data.csv" for official data
```

- “*employment_realGDP_data.csv*: provides provincial real GDP per worker in agriculture and nonagriculture in 2000 and 2005. Also provides employment in agriculture and nonagriculture in 2000 and 2005.”

```
1 [N vars]=size(data.textdata); % data structure: 31*9
2 region=data.textdata(2:N,1);
3 rYLa_data=data.data(:,1);
4 rYLn_data=data.data(:,2);
5 La_data=data.data(:,3);
6 Ln_data=data.data(:,4); % real GDP per worker in each sector
7 l=(La_data+Ln_data);
8 L(1:N,1)=sum(La_data+Ln_data); % different from the paper: 629 million
9 rYL_data=(rYLa_data.*La_data+rYLn_data.*Ln_data)./(La_data+Ln_data); % ...
   average real GDP per worker
10 mu=[rYLa_data rYLn_data]'; % 30*2
11 mu=mu(:); % 60*1
12 rYLa_data2005=data.data(:,5);
13 rYLn_data2005=data.data(:,6);
14 drYLa_data=data.data(:,7);
15 drYLn_data=data.data(:,8);
16 drYL_data=[drYLa_data drYLn_data]';
17 drYL_data=drYL_data(:);
18
19 % International values
20 L(N,1)=2102.978587;
```

```
1 % Load Initial Trade Shares
2 pi_ag=importdata('trade_ag.csv'); % 31*31
3 pi_ag=pi_ag./repmat(sum(pi_ag,2),1,N); % normalization (unnecessary?)
```

```

4 pi_na=importdata('trade_na.csv');
5 pi_na=pi_na./repmat(sum(pi_na,2),1,N);

```

- “trade_ag.csv and trade_na.csv: the bilateral trade matrix for agriculture goods and for non-agricultural goods between provinces in 2002.”

```

1 % Load Changes in Trade Costs
2 data=importdata('tauhat.csv');
3 dni_ag_measured=reshape(data.data(:,1),N,N)';
4 dni_na_measured=reshape(data.data(:,2),N,N)'; % symmetric (formula ...
   missing)
5 dni_asym_ag=reshape(data.data(:,3),N,N)';
6 dni_asym_na=reshape(data.data(:,4),N,N)'; % asymmetric (formula missing)

```

- “tauhat.csv: the measured change in trade costs.” (Typo in the ReadMe file.)

```

1 % Load Migration Shares
2 global cij L0 mij2000
3 data=importdata('mij2000.csv');
4 mij2000=reshape(data.data(:,2),60,60)'; % each row is a ...
   province-industry pair
5 data=importdata('mij2005.csv');
6 mij2005=reshape(data.data(:,2),60,60)'; % each row is a ...
   province-industry pair

```

- “mij2000.csv and mij2005.csv: the bilateral migration shares and non-migrant shares for each pair of province-sector (ag/nonag) pairs.”

```

1 % Initial Employment Vector
2 L1=[La_data Ln_data]';
3 L1=L1(:);
4
5 % Hukou Registrations, by Province
6 global La Ln
7 L0=(L1'*inv(mij2000))';
8 reshape(L1',2,N-1)';
9 La=ans(:,1);
10 Ln=ans(:,2);

```

- Hukou registration is calculated rather than observed (due to the fact that census is conducted based on residence).
- $L1$ is the employment vector (after migration). $L1$ & $L0$ satisfy

$$L1' = L0' \cdot mij2000.$$

Therefore,

$$L0 = (L1' \cdot mij2000^{-1})'.$$

```

1      % Production Function Input Shares
2      beta_ag=0.287;
3      beta_na=0.219; % These reflect Brandt, Restuccia, and Adamopolous's ...
4      % paper for ag's VA shares + the IO data
5      eta_ag=0.277;
6      eta_na=0.025;
7
8      % Intermediate Input Shares
9      gamma=[0.397 0.603;
10             0.063 0.937]; % sigma in the paper (typo in Table 3)
11
12     % Common Parameters
13     alpha=1-0.13; % set to 1 for no housing; otherwise 1-0.13
14     kappa=1.5;
15     theta=4;
16     epsilon=0.09545; % agriculture's share of total end-use in the IO tables
17     % psi in the paper

```

```

1      % Initial total nominal expenditures and income (see write-up)
2      global X_ag X_na R_ag R_na I Ir Iu omega_ag omega_ag2 omega_na2 epsilon
3      I0=zeros(N,1);
4      I=ones(N,1);
5      X0_ag=zeros(N,1);
6      X_ag=ones(N,1);
7      X0_na=zeros(N,1);
8      X_na=ones(N,1);
9      while sum((X0_ag-X_ag).^2)+sum((X0_na-X_na).^2)>1e-20
10         X0_ag=X_ag;
11         X0_na=X_na;
12         R_ag=pi_ag'*X0_ag;
13         R_na=pi_na'*X0_na;
14
15         Ir=(beta_ag+eta_ag).*R_ag./alpha; % Rural Income
16         Iu=(beta_na+eta_na).*R_na./alpha; % Urban Income
17         I=Ir+Iu; % Total Provincial Income
18         Ir=Ir./sum(I);
19         Iu=Iu./sum(I); % Normalization
20
21         Da_r=alpha*epsilon.*Ir;
22         Da_u=alpha*epsilon.*Iu;
23         Dn_r=alpha.*(1-epsilon).*Ir;
24         Dn_u=alpha.*(1-epsilon).*Iu;
25         Ds_r=(1-alpha).*Ir;
26         Ds_u=(1-alpha).*Iu;
27
28         X_ag=(Da_r+Da_u+(1-beta_ag-eta_ag).*gamma(1,1).*R_ag
29              +(1-beta_na-eta_na).*gamma(2,1).*R_na);
30         X_na=(Dn_r+Dn_u+(1-beta_na-eta_na).*gamma(2,2).*R_na
31              +(1-beta_ag-eta_ag).*gamma(1,2).*R_ag);
32     end

```

- Total revenue:

$$R_n^j = \sum_{i=1}^{N+1} \pi_{in}^j X_i^j. \quad (6)$$

- Derivation of income:

$$\begin{aligned} r_n^j \bar{S}_n^j &= (1 - \alpha) v_n^j L_n^j + \eta^j R_n^j; \\ v_n^j L_n^j &= (1 - \alpha) v_n^j L_n^j + \eta^j R_n^j + w_n^j L_n^j; \\ v_n^j L_n^j &= (\eta^j R_n^j + \beta^j R_n^j) / \alpha. \end{aligned}$$

- The final demand for good j in region n :

$$D_n^j = \alpha \psi^j \sum_{k \in \{ag, na\}} v_n^k L_n^k. \quad (2)$$

- The final demand for housing:

$$(1 - \alpha) \sum_{k \in \{ag, na\}} v_n^k L_n^k.$$

- Total expenditure:

$$X_n^j = D_n^j + \sum_k \sigma^{kj} R_n^k. \quad (7)$$

- Gauss-Seidel iteration ($\Rightarrow$ The iteration process calculates X_n^j without solving for wages, or the wage is implicitly solved during the process):

1. Initial guess: $[X_n^j]^0 = 1, \forall n, j$;
2. Calculate $[\tilde{X}_n^j]^1$ based on parameters and data on π_{ni}^j ;
3. Iteration process:
 - $[X_n^j]^{i+1} = \omega \cdot [\tilde{X}_n^j]^{i+1} + (1 - \omega)[X_n^j]^i$;
 - Let $\omega = 1$.
4. Iterate until convergence ($RSS < 10^{-20}$).

```

1      % Solve for initial and 2005 real income per worker
2      global Vi dVi_data
3      Via(1:N-1,1)=rYLa_data; % real income ag, 2000
4      Vin(1:N-1,1)=rYLn_data; % real income na, 2000
5      Via2005(1:N-1,1)=rYLa_data.*drYLa_data; % real income ag, 2005
6      Vin2005(1:N-1,1)=rYLn_data.*drYLn_data; % real income na, 2005
7      % so drYL is the relative change of real income
8      Vi=[Via Vin]';
9      Vi=Vi(:); % 60*1
10     Vi2005=[Via2005 Vin2005]';
11     Vi2005=Vi2005(:);
12     dVi_data=[Vi2005./Vi;1.055;1.055]; % the relative changes in the ...
        international sectors are set to 1.055
13     reshape(dVi_data',2,N)';
14     dVa_data=ans(:,1);
15     dVn_data=ans(:,2);

```

```

1 % Solve for initial migration costs (per effective labour)
2 Vimat= repmat(Vi,1,2*(N-1)); % 60*60, each row represents a region
3 Vjmat= repmat(Vi',2*(N-1),1); % 60*60, each column represents a region
4 cij=(mij2000./ repmat(diag(mij2000),1,2*(N-1))).^(1/kappa)./(Vjmat./Vimat); ...
    % land cost seems to be neglected here
5 cij2000=cij; % redundant actually
6 Vimat2005= repmat(Vi2005,1,2*(N-1));
7 Vjmat2005= repmat(Vi2005',2*(N-1),1);
8 cij2005=(mij2005./ repmat(diag(mij2005),1,2*(N-1))).^(1/kappa)
9 ./(Vjmat2005./Vimat2005);
10 mij=(cij2000.*Vjmat).^kappa;
11 mij=mij./(sum(mij,2)*ones(1,2*(N-1)));
12 cijhat=ones(2*(N-1),2*(N-1));
13 L1base=(L0'*mij)'; % same as L1
14 L2005=(L0'*mij2005)';

```

- The land opportunity cost δ_{ni}^{jk} seems to be first neglected here. In the calculation above, migration won't affect the potential income of migrants.
- From Eqn.10, we have

$$m_{ni}^{jk} = \frac{(V_i^k \delta_{ni}^{jk} / \mu_{ni}^{jk})^\kappa}{\sum_{k'} \sum_{i'} (V_{i'}^{k'} \delta_{ni'}^{jk'} / \mu_{ni'}^{jk'})^\kappa}. \quad (10)$$

Neglecting the land opportunity cost, we have,

$$m_{ni}^{jk} = \frac{(V_i^k / \mu_{ni}^{jk})^\kappa}{\sum_{k'} \sum_{i'} (V_{i'}^{k'} / \mu_{ni'}^{jk'})^\kappa}.$$

Therefore,

$$\frac{m_{ni}^{jk}}{m_{nn}^{jj}} = \frac{(V_i^k / \mu_{ni}^{jk})^\kappa}{(V_n^j / \mu_{nn}^{jj})^\kappa} = \left(\frac{V_i^k / \mu_{ni}^{jk}}{V_n^j} \right)^\kappa.$$

- We can see that cij is the reciprocal of the migration utility cost:

$$\frac{1}{\mu_{ni}^{jk}} = \left(\frac{m_{ni}^{jk}}{m_{nn}^{jj}} \right)^{\frac{1}{\kappa}} \left(\frac{V_n^j}{V_i^k} \right).$$

- Then, mij is calculated according to Eqn.10 (still neglecting the land opportunity cost).
- $L1base$ is the same matrix as $L1$, why?
 - It is because mij and $mij2000$ are the same.
 - $L1$ is directly observed from data. $L0$ is derived using $mij2000$ & $L1$.
 - We use $mij2000$ (from data) to derive cij , then use cij to derive mij . The two derivation processes use the same formula, Eqn.10. Therefore, mij and $mij2000$ end up the same.

$$L0 = (L1' \cdot mij2000^{-1})' \& L1base = (L0' \cdot mij) \Rightarrow L1 = L1base.$$

```

1 % Land Rebate Adjustments to Migration Costs
2 global Cnnjj Cnnjj2005 Cnijk Cnijk2005
3 M_mL_term=Libase./(diag(mij).*L0); % 60*1
4 temp=reshape(M_mL_term,2,N-1)'; % 30*2
5 M_mL_term_ag=temp(:,1);
6 M_mL_term_na=temp(:,2);
7 Cnnjj_ag=1+(eta_ag+(1-alpha)*beta_ag).*M_mL_term_ag./(alpha.*beta_ag);
8 Cnnjj_na=1+(eta_na+(1-alpha)*beta_na).*M_mL_term_na./(alpha.*beta_na);
9 Cnnjj=[Cnnjj_ag Cnnjj_na]';
10 Cnnjj=Cnnjj(:);
11 Cnijk=(1-eye(2*(N-1),2*(N-1))).* repmat(Cnnjj,1,2*(N-1)).*cij;
12 Cnijk2000=Cnijk;
13 Cnnjj2000=Cnnjj; % also redundant
14
15 % As above, but for 2005
16 M_mL_term2005=L2005./(diag(mij2005).*L0);
17 temp=reshape(M_mL_term2005,2,N-1)';
18 M_mL_term_ag2005=temp(:,1);
19 M_mL_term_na2005=temp(:,2);
20 Cnnjj_ag2005=1+(eta_ag+(1-alpha)*beta_ag).*
21     M_mL_term_ag2005./(alpha.*beta_ag);
22 Cnnjj_na2005=1+(eta_na+(1-alpha)*beta_na).*
23     M_mL_term_na2005./(alpha.*beta_na);
24 Cnnjj2005=[Cnnjj_ag2005 Cnnjj_na2005]';
25 Cnnjj2005=Cnnjj2005(:);
26 Cnijk2005=(1-eye(2*(N-1),2*(N-1))).* repmat(Cnnjj2005,1,2*(N-1)).*cij2005;

```

- $Cnnjj$ is the land opportunity cost term:

$$\delta_{ni}^{jk} = \begin{cases} 1 + \left(\frac{(1-\alpha)\beta^j + \eta^j}{\alpha\beta^j} \right) \frac{L_n^j}{L_{nn}^{jj}}, & \text{if } n = i \text{ and } j = k; \\ 1, & \text{if } n \neq i \text{ or } j \neq k. \end{cases} \quad (9)$$

- Why does it have to be calculated separately (agricultural & non-agricultural)?
* β^j is different in different sectors, and it is stored separately.

- $Cnijk$:

- $eye()$ generates an identity matrix. $1 - eye()$ generates a matrix where diagonal elements are 0 and off-diagonal elements are 1.
- Therefore,

$$Cnijk(i, j) = \begin{cases} 0, & \text{if } i = j; \\ Cnnjj(i) \times cij(i, j), & \text{if } i \neq j. \end{cases}$$

- The non-zero elements of $Cnijk$ represent $\delta_{nn}^{jj}/\mu_{ni}^{jk}$.

```

1 % International initial labour distribution
2 Ln_data(N,1)=L(N)./(1+(beta_ag./beta_na).*(R_ag(N)./R_na(N))); % how?
3 La_data(N,1)=L(N)-Ln_data(N);

```

- According to Cobb-Douglas production technologies, we have,

$$\beta^j R_w^j = w_w^j L_w^j.$$

Therefore,

$$(\beta^{ag} R_w^{ag})/(\beta^{na} R_w^{na}) = (w_w^{ag} L_w^{ag})/(w_w^{na} L_w^{na}).$$

- The question is, how is $L_w^{na} = L_w[1 + (w_w^{ag} L_w^{ag})/(w_w^{na} L_w^{na})]^{-1}$?

$$\begin{aligned} \text{RHS} &= w_w^{na} L_w^{na} L_w (w_w^{na} L_w^{na} + w_w^{ag} L_w^{ag})^{-1} \\ &= L_w^{na} (L_w^{na} + L_w^{ag}) (L_w^{na} + \frac{w_w^{ag}}{w_w^{na}} L_w^{ag})^{-1}. \end{aligned}$$

- The authors have implicitly assume that $w_w^{na} = w_w^{ag}$.
- Why so? Walras law.

```
1 % Average symmetric trade costs
2 tau_ag=((diag(pi_ag)*diag(pi_ag'))./(pi_ag.*pi_ag')).^(1./(2.*theta));
3 tau_na=((diag(pi_na)*diag(pi_na'))./(pi_na.*pi_na')).^(1./(2.*theta));
```

- Eqn.15 gives the measure of symmetric trade cost:

$$\bar{\tau}_{ni}^j \equiv \sqrt{\pi_{ni}^j \pi_{in}^j} = \left(\frac{\pi_{nn}^j \pi_{ii}^j}{\pi_{ni}^j \pi_{in}^j} \right)^{\frac{1}{2\theta}}. \quad (15)$$

```
1 %% Get the Initial Equilibrium of the Model
2 global dTa dTn
3 dTa=ones(N,1);
4 dTn=ones(N,1);
5 dni_ag=ones(N,N);
6 dni_na=ones(N,N);
7 new=fsolve(@(X)main_simulate(X),ones(6*N,1),optimset('Display','off'));
8 dwa=new(1:N);
9 dwn=new(N+1:2*N);
10 dPa=new(2*N+1:3*N);
11 dPn=new(3*N+1:4*N);
12 dLa=new(4*N+1:5*N);
13 dLn=new(5*N+1:6*N);
14 dL=[dLa dLn]'; dL=dL(:);
15 dw=[dwa dwn]'; dw=dw(:);
```

- *fsolve* function is used to solve systems of nonlinear equations. It is part of the *Optimization Toolbox*.
- Basic syntax of *fsolve* function:

```
1 [x, fval, exitflag, output, jacobian] = fsolve(fun, x0, options)
```

- *x*: the solution found;
- *fval*: function value at the solution;

- *exitflag*: value describing the exiting condition;
 - *output*: structure containing optimization process information;
 - *jacobian*: Jacobian matrix at the solution;
 - *fun*: the function to solve;
 - *x0*: initial guess;
 - *options*: optimization options.
- Default algorithm of *fsolve*: *Trust-Region Dogleg Method*.
 - It approximates the nonlinear system locally using a linear or quadratic model.
 - It restricts the step size to a trust region where the model is deemed reliable.
 - It selects a step along a dogleg path: a trajectory balancing
 - * *Newton's method* (fast local convergence but potentially unstable);
 - * *Steepest descent* ("global" convergence but slow).
 - Requirement:
 - * default for systems with the same number of equations and variables;
 - * requires the Jacobian matrix (either provided by users or can be approximated via finite difference).
 - Translate the nonlinear system into a minimization problems, for example:

$$F(x) = 0 \Rightarrow \min_x \phi(x) = \frac{1}{2} \|F(x)\|^2.$$

- *Newton's method*:

$$\begin{aligned}\phi(x_{t+1}) &\approx \phi(x_t) + (x_{t+1} - x_t)' \nabla \phi(x_t) + \frac{1}{2} (x_{t+1} - x_t)' H_t (x_{t+1} - x_t); \\ \frac{\partial \phi(x_{t+1})}{\partial x_{t+1}} &= \nabla \phi(x_t) + H_t (x_{t+1} - x_t) = 0; \\ p_{Newton} &= -H_t^{-1} \nabla \phi(x_t).\end{aligned}$$

- * Unstable? *Newton's method* requires $\phi(x_t)$ to be locally convex.

$$\begin{aligned}\phi(x_{t+1}) &\approx \phi(x_t) + (x_{t+1} - x_t)' \nabla \phi(x_t) \\ &= \phi(x_t) + (-H_t^{-1} \nabla \phi(x_t))' \nabla \phi(x_t) \\ &= \phi(x_t) + \nabla \phi(x_t)' (-H_t^{-1}) \nabla \phi(x_t).\end{aligned}$$

$$\phi(x_{t+1}) - \phi(x_t) < 0 \Rightarrow \nabla \phi(x_t)' (-H_t^{-1}) \nabla \phi(x_t) < 0.$$

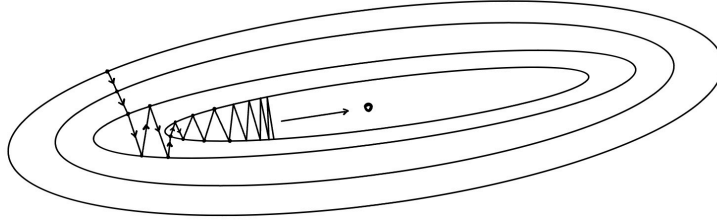
- *Steepest descent*:

$$p_{SD} = -\nabla \phi(x_t).$$

- * p_{SD} only tells the direction of the next step. *Steepest descent* tells us that the direction of the negative gradient is the direction in which the function value decreases the fastest. Why?

$$\begin{aligned}\phi(x_{t+1}) &\approx \phi(x_t) + (x_{t+1} - x_t)' \nabla \phi(x_t); \\ \mathcal{L} &= -\phi(x_t) - (x_{t+1} - x_t)' \nabla \phi(x_t) - \lambda[(x_{t+1} - x_t)'(x_{t+1} - x_t) - r]; \\ \frac{\partial \mathcal{L}}{\partial x_{t+1}} &= -\nabla \phi(x_t) - 2\lambda(x_{t+1} - x_t) = 0; \\ (x_{t+1} - x_t) &= -\frac{1}{2\lambda} \nabla \phi(x_t).\end{aligned}$$

- * Why is *Steepest descent* slow? Narrow valley problem.



– Iteration steps for *Trust-Region Dogleg Method*:

1. Choose initial point x_0 , trust-region radius Δ_0 , and tolerance level ϵ .
 2. Compute the Jacobian, then p_{Newton} & p_{SD} .
 3. If $\|p_{Newton}\| \leq \Delta_t$, then choose $p_t = p_{Newton}$, otherwise choose $p_t = \beta p_{Newton} + (1 - \beta)p_{SD}$ where β is chosen to let $\|p_t\| = \Delta_t$.
 4. Obtain a new point and iterate the step 2-3. Δ_t can be expanded if the prediction is accurate, shrunk otherwise.
 5. Stop when $\|\nabla \phi(x_t)\| < \epsilon$.
- Functions to solve: main_simulate.m.
 - Variables to solve: $dwa, dwn, dPa, dPn, dLa, dLn$.
 - dxx here represents the hat algebra in the article. Therefore, $dwa = \hat{w}^{ag} = w^{ag}/w^{ag}$, and others are similar.

```

1  % New sectoral revenues
2  R_ag_new=dwa.*dLa.*R_ag;
3  R_na_new=dwn.*dLn.*R_na;
4
5  % New household final demand spending
6  Ir_new=(beta_ag+eta_ag).*R_ag_new./alpha;
7  dIr=Ir_new./Ir;
8  Iu_new=(beta_na+eta_na).*R_na_new./alpha;
9  dIu=Iu_new./Iu;
10 Da_new_r=alpha*epsilon.*Ir_new;
11 Da_new_u=alpha*epsilon.*Iu_new;
12 Dn_new_r=(alpha*(1-epsilon)).*Ir_new;
```

```

13 Dn_new_u=(alpha*(1-epsilon)).*Iu_new;
14 Ds_new_r=(1-alpha).*Ir_new;
15 Ds_new_u=(1-alpha).*Iu_new;
16
17 % Change in land prices
18 dra(:,1)=dwa.*dLa;
19 drn(:,1)=dwn.*dLn;
20
21 % New trade shares for agriculture and nonagriculture
22 inside_ag=dwa'.^beta_ag
23     .*(dPa'.^gamma(1,1).*dPn'.^gamma(1,2)).^(1-beta_ag-eta_ag)
24     .*dra'.^eta_ag./(dTa'.^(1./theta));
25 denominator_ag=
26     sum(pi_ag.*(dni_ag.*kron(inside_ag,ones(N,1))).^(-theta),2);
27 pi_ag_new=(pi_ag.*(dni_ag.*kron(inside_ag,ones(N,1))).^(-theta))
28     ./kron(denominator_ag,ones(1,N));
29 inside_na=dwn'.^beta_na
30     .*(dPa'.^gamma(2,1).*dPn'.^gamma(2,2)).^(1-beta_na-eta_na)
31     .*drn'.^eta_na./(dTn'.^(1./theta));
32 denominator_na
33     =sum(pi_na.*(dni_na.*kron(inside_na,ones(N,1))).^(-theta),2);
34 pi_na_new=(pi_na.*(dni_na.*kron(inside_na,ones(N,1))).^(-theta))
35     ./kron(denominator_na,ones(1,N));
36
37 % Sector expenditures
38 X_ag_new=inv(pi_ag_new')*R_ag_new;
39 X_na_new=inv(pi_na_new')*R_na_new;
40
41 % Change in real income per effective labour, by sector
42 dP=dPa.^epsilon.*dPn.^(1-epsilon);
43 dVa=dwa./(dP.^alpha.*dra.^(1-alpha));
44 dVn=dwn./(dP.^alpha.*drn.^(1-alpha));
45 dV=[dVa dVn]'; dV=dV(:);
46 dPa_vector=[dPa dPa]'; dPa_vector=dPa_vector(:);
47 dv_vector=[dIr./dLa dIu./dLn]'; dv_vector=dv_vector(:);
48
49 % Migration Shares
50 mnn1=diag(mij2000); mnn0=ones(2*(N-1),1);
51 while sum((mnn0-mnn1).^2)>1e-20
52     mnn0=mnn1;
53     M_mL_term=(L1base.*dL(1:2*(N-1),1))./(mnn0.*L0);
54     temp=reshape(M_mL_term,2,N-1)';
55     M_mL_term_ag=temp(:,1);
56     M_mL_term_na=temp(:,2);
57     Cnnjj_ag=1+(eta_ag+(1-alpha)*beta_ag).*M_mL_term_ag./(alpha.*beta_ag);
58     Cnnjj_na=1+(eta_na+(1-alpha)*beta_na).*M_mL_term_na./(alpha.*beta_na);
59     Cnnjj=[Cnnjj_ag Cnnjj_na]'; Cnnjj=Cnnjj(:);
60     cij=Cnij+k*eye(2*(N-1),2*(N-1)).*repmat(Cnnjj,1,2*(N-1));
61     Vjmat=repmat((dV(1:2*(N-1)).*Vi)',2*(N-1),1);
62     mij=(cij.*Vjmat).^kappa;
63     mij=mij./(sum(mij,2)*ones(1,2*(N-1)));
64     mnn1=mean([diag(mij) mnn0],2);
65 end

```

```

1 % Initial Welfare Vector
2 global omega_welfare

```

```

3      omega_welfare=(L0./sum(L0)).*Vi.*Cnnjj2000.*diag(mij).^(-1/kappa)
4      ./sum((L0./sum(L0)).*Vi.*Cnnjj2000.*diag(mij).^(-1/kappa));

```

- Aggregate welfare:

$$W = \sum_n \sum_j \frac{\bar{L}_n^j}{\sum_n \sum_j \bar{L}_n^j} V_n^j \delta_{nn}^{jj} (m_{nn}^{jj})^{-1/\kappa}.$$

- We can see that the change in aggregate welfare is

$$\hat{W} = \sum_n \sum_j \omega_n^j \hat{V}_n^j \hat{\delta}_{nn}^{jj} (\hat{m}_{nn}^{jj})^{-1/\kappa}, \quad (11)$$

where $\omega_n^j = \frac{\bar{L}_n^j}{\sum_n \sum_j \bar{L}_n^j} V_n^j \delta_{nn}^{jj} (m_{nn}^{jj})^{-1/\kappa} / W$.

2 main_simulate.m

```

1 function F=main_simulate(X)
2
3 global N theta beta_ag beta_na eta_ag eta_na dni_ag dni_na pi_ag pi_na ...
   L0 kappa alpha dTa dTn Vi mij2000
4 global R_ag R_na Ir Iu epsilon La_data Ln_data L1base gamma Cnijk
5
6 dwa=X(1:N);
7 dwn=X(N+1:2*N);
8 dPa=X(2*N+1:3*N);
9 dPn=X(3*N+1:4*N);
10 dLa=X(4*N+1:5*N);
11 dLn=X(5*N+1:6*N);
12 dL=[dLa dLn]';
13 dL=dL(:);
14 dw=[dwa dwn]';
15 dw=dw(:);

```

```

1 % New sectoral revenues
2 R_ag_new=dwa.*dLa.*R_ag;
3 R_na_new=dwn.*dLn.*R_na;

```

$$\begin{aligned}
 R_n^j &= (\beta^j)^{-1} w_n^j L_n^j; \\
 R_n^{j'} &= (\beta^j)^{-1} w_n^{j'} L_n^{j'}; \\
 R_n^{j'} &= \frac{w_n^{j'}}{w_n^j} \frac{L_n^{j'}}{L_n^j} R_n^j = \hat{w}_n^j \hat{L}_n^j R_n^j.
 \end{aligned}$$

```

1 % New household final demand spending
2 Ir_new=(beta_ag+eta_ag).*(R_ag_new./alpha; % same equation, but with new ...
   revenue
3 dIr=Ir_new./Ir;
4 Iu_new=(beta_na+eta_na).*(R_na_new./alpha;
5 dIu=Iu_new./Iu;
6 Da_new_r=alpha*epsilon.*Ir_new;
7 Da_new_u=alpha*epsilon.*Iu_new;
8 Dn_new_r=(alpha*(1-epsilon)).*Ir_new;
9 Dn_new_u=(alpha*(1-epsilon)).*Iu_new;
10 Ds_new_r=(1-alpha).*Ir_new;
11 Ds_new_u=(1-alpha).*Iu_new; % same equation, but with new income

```

```

1 % Change in land prices
2 dra(:,1)=dwa.*dLa;
3 drn(:,1)=dwn.*dLn;

```

- From Eqn.8, we have the total land income:

$$r_n^j \bar{S}_n^j = \left[\frac{(1-\alpha)\beta^j + \eta^j}{\alpha\beta^j} \right] w_n^j L_n^j. \quad (8)$$

- Therefore, the new total land income after the changes is,

$$r_n^{j'} \bar{S}_n^{j'} = \left[\frac{(1-\alpha)\beta^j + \eta^j}{\alpha\beta^j} \right] w_n^{j'} L_n^{j'}.$$

- Finally, we have,

$$\hat{r}_n^j = \hat{w}_n^j \hat{L}_n^j.$$

```

1  % New trade shares for agriculture and nonagriculture
2  inside_ag=dwa'.^beta_ag
3  .*(dPa'.^gamma(1,1).*dPn'.^gamma(1,2)).^(1-beta_ag-eta_ag) % gamma ...
   represents sigma in the article (before normalization)
4  .*dra'.^eta_ag./(dTa'.^(1./theta));
5  denominator_ag=
6  sum(pi_ag.*(dni_ag.*kron(inside_ag,ones(N,1))).^(-theta),2);
7  pi_ag_new=(pi_ag.*(dni_ag.*kron(inside_ag,ones(N,1))).^(-theta))
8  ./kron(denominator_ag,ones(1,N));
9
10 inside_na=dwn'.^beta_na
11 .*(dPa'.^gamma(2,1).*dPn'.^gamma(2,2)).^(1-beta_na-eta_na)
12 .*drn'.^eta_na./(dTn'.^(1./theta));
13 denominator_na=
14 sum(pi_na.*(dni_na.*kron(inside_na,ones(N,1))).^(-theta),2);
15 pi_na_new=(pi_na.*(dni_na.*kron(inside_na,ones(N,1))).^(-theta))
16 ./kron(denominator_na,ones(1,N));

```

- The relative changes in the cost function (times productivity):

$$\begin{aligned}
c_n^j &= (w_n^j)^{\beta^j} (r_n^j)^{\eta^j} (P_n^{ag})^{\sigma^{j,ag}} (P_n^{na})^{\sigma^{j,na}}; \\
\hat{c}_n^j &= (\hat{w}_n^j)^{\beta^j} (\hat{r}_n^j)^{\eta^j} (\hat{P}_n^{ag})^{\sigma^{j,ag}} (\hat{P}_n^{na})^{\sigma^{j,na}}.
\end{aligned} \tag{18}$$

- The relative changes in the trade share:

$$\begin{aligned}
\pi_{ni}^j &= \frac{T_i^j (\tau_{ni}^j c_i^j)^{-\theta}}{\sum_{m=1}^{N+1} T_m^j (\tau_{nm}^j c_m^j)^{-\theta}}; \\
\pi_{ni}^{j'} &= \frac{T_i^{j'} (\tau_{ni}^{j'} c_i^{j'})^{-\theta}}{\sum_{m=1}^{N+1} T_m^{j'} (\tau_{nm}^{j'} c_m^{j'})^{-\theta}}; \\
\hat{\pi}_{ni}^j &= \left[\frac{T_i^j (\tau_{ni}^j c_i^j)^{-\theta}}{\sum_{m=1}^{N+1} T_m^j (\tau_{nm}^j c_m^j)^{-\theta}} \right]^{-1} \frac{T_i^{j'} (\tau_{ni}^{j'} c_i^{j'})^{-\theta}}{\sum_{m=1}^{N+1} T_m^{j'} (\tau_{nm}^{j'} c_m^{j'})^{-\theta}} \\
&= \frac{\hat{T}_i^j (\hat{\tau}_{ni}^j \hat{c}_i^j)^{-\theta}}{\sum_{m=1}^{N+1} \frac{T_m^{j'} (\tau_{nm}^{j'} c_m^{j'})^{-\theta}}{\sum_{m=1}^{N+1} T_m^j (\tau_{nm}^j c_m^j)^{-\theta}}} \\
&= \frac{\hat{T}_i^j (\hat{\tau}_{ni}^j \hat{c}_i^j)^{-\theta}}{\sum_{m=1}^{N+1} \frac{T_m^j (\tau_{nm}^j c_m^j)^{-\theta}}{\sum_{m=1}^{N+1} T_m^j (\tau_{nm}^j c_m^j)^{-\theta}} \hat{T}_m^j (\hat{\tau}_{nm}^j \hat{c}_m^j)^{-\theta}} \\
&= \frac{\hat{T}_i^j (\hat{\tau}_{ni}^j \hat{c}_i^j)^{-\theta}}{\sum_{m=1}^{N+1} \pi_{nm}^j \hat{T}_m^j (\hat{\tau}_{nm}^j \hat{c}_m^j)^{-\theta}}.
\end{aligned} \tag{19}$$

- Code correspondence:

$$inside_i^j = (w_i^j)^{\beta^j} (r_i^j)^{\eta^j} (P_i^{ag})^{\sigma^{j,ag}} (P_i^{na})^{\sigma^{j,na}} / (\hat{T}_i^j)^{\frac{1}{\theta}} = \hat{c}_i^j / (\hat{T}_i^j)^{\frac{1}{\theta}}.$$

$$denominator_n^j = \sum_{m=1}^{N+1} \pi_{nm}^j (\hat{\tau}_{nm}^j \hat{c}_m^j / (\hat{T}_m^j)^{\frac{1}{\theta}})^{-\theta} = \sum_{m=1}^{N+1} \pi_{nm}^j \hat{T}_m^j (\hat{\tau}_{nm}^j \hat{c}_m^j)^{-\theta}.$$

$$\pi_{ni}^{j'} = \pi_{ni}^j (\hat{\tau}_{ni}^j inside_i^j)^{-\theta} / denominator_n^j = \pi_{ni}^j \frac{\hat{T}_i^j (\hat{\tau}_{ni}^j \hat{c}_i^j)^{-\theta}}{\sum_{m=1}^{N+1} \pi_{nm}^j \hat{T}_m^j (\hat{\tau}_{nm}^j \hat{c}_m^j)^{-\theta}}.$$

```

1      % Sector expenditures
2      X_ag_new=inv(pi_ag_new')*R_ag_new;
3      X_na_new=inv(pi_na_new')*R_na_new;

```

- Recall that total revenue:

$$R_n^j = \sum_{i=1}^{N+1} \pi_{in}^j X_i^j. \quad (6)$$

- In a matrix form:

$$R_ag_new = pi_ag_new \cdot X_ag_new.$$

- Therefore,

$$X_ag_new = (pi_ag_new)^{-1} \cdot R_ag_new.$$

```

1      % Change in real income per effective labour, by sector
2      dP=dPa.^epsilon.*dPn.^(1-epsilon);
3      dVa=dwa./(dP.^alpha.*dra.^(1-alpha));
4      dVn=dwn./(dP.^alpha.*drn.^(1-alpha));
5      dV=[dVa dVn]';
6      dV=dV(:);
7      dPa_vector=[dPa dPa]';
8      dPa_vector=dPa_vector(:);
9      dv_vector=[dIr./dLa dIu./dLn]';
10     dv_vector=dv_vector(:);

```

- The Cobb-Douglas utility of the worker:

$$u_n^j = \varepsilon_n^j [(C_n^{j,ag})^{\psi^{ag}} (C_n^{j,na})^{\psi^{na}}]^{\alpha} (S_n^{j,h})^{1-\alpha}. \quad (1)$$

- Consumer price index according to the C-D utility is

$$[(P_n^{j,ag})^{\psi^{ag}} (P_n^{j,na})^{\psi^{na}}]^{\alpha} (r_n^j)^{1-\alpha}.$$

- Real income of the worker:

$$V_n^j = \frac{w_n^j}{[(P_n^{j,ag})^{\psi^{ag}} (P_n^{j,na})^{\psi^{na}}]^{\alpha} (r_n^j)^{1-\alpha}}.$$

- The change in real income is

$$\hat{V}_n^j = \frac{\hat{w}_n^j}{[(\hat{P}_n^{j,ag})^{\psi^{ag}} (\hat{P}_n^{j,na})^{\psi^{na}}]^{\alpha} (\hat{r}_n^j)^{1-\alpha}}. \quad (21)$$

– Typo in the online appendix.

```

1  % Migration Shares
2  mnn1=diag(mij2000);
3  mnn0=ones(2*(N-1),1);
4  while sum((mnn0-mnn1).^2)>1e-20
5      mnn0=mnn1;
6      M_mL_term=(L1base.*dL(1:2*(N-1),1))./(mnn0.*L0);
7      temp=reshape(M_mL_term,2,N-1)';
8      M_mL_term_ag=temp(:,1);
9      M_mL_term_na=temp(:,2);
10     Cnnjj_ag=1+(eta_ag+(1-alpha)*beta_ag).*M_mL_term_ag./(alpha.*beta_ag);
11     Cnnjj_na=1+(eta_na+(1-alpha)*beta_na).*M_mL_term_na./(alpha.*beta_na);
12     Cnnjj=[Cnnjj_ag Cnnjj_na]';
13     Cnnjj=Cnnjj(:);
14     cij=Cnij+eye(2*(N-1),2*(N-1)).* repmat(Cnnjj,1,2*(N-1));
15     Vjmat=repmat((dV(1:2*(N-1)).*Vi)',2*(N-1),1);
16     mij=(cij.*Vjmat).^kappa;
17     mij=mij./(sum(mij,2)*ones(1,2*(N-1)));
18     mnn1=mean([diag(mij) mnn0],2);
19 end

```

- Gauss-Seidel iteration to calculate the migration shares.
- Possible mistake here?
 - $Cnij$: a matrix where diagonal elements are 0, while non-diagonal elements represent $\delta_{nn}^{jj}/\mu_{ni}^{jk}$.
 - cij : a matrix where diagonal elements represent δ_{nn}^{jj} , while non-diagonal elements represent $\delta_{nn}^{jj}/\mu_{ni}^{jk}$.
 - However, cij is supposed to be a matrix where diagonal elements represent δ_{nn}^{jj} (since $\mu_{nn}^{jj} = 1$), while non-diagonal elements represent $1/\mu_{ni}^{jk}$ (since $\delta_{ni}^{jk} = 1$).

```

1  % New labour allocation
2  L1=(L0'*mij)';

```

```

1  % The system of equations
2  F=[
3
4      dPa-denominator_ag.^(-1/theta);
5      dPn-denominator_na.^(-1/theta);
6
7      L1(1:2*(N-1))./L1base(1:2*(N-1))-dL(1:2*(N-1));
8      dLa(N).*La_data(N)+dLn(N).*Ln_data(N)-(La_data(N)+Ln_data(N)); % ...
          labor market clearing
9      dwa(N)-dwn(N); % normalization due to Walras law
10
11     X_ag_new-(Da_new_r+Da_new_u+(1-beta_ag-eta_ag).*gamma(1,1)
12             .*R_ag_new+(1-beta_na-eta_na).*gamma(2,1).*R_na_new);
13     X_na_new-(Dn_new_r+Dn_new_u+(1-beta_na-eta_na).*gamma(2,2)
14             .*R_na_new+(1-beta_ag-eta_ag).*gamma(1,2).*R_ag_new);
15
16 ];

```

- The change of price index.

- From Eqn.5, we have the price index:

$$P_n^j = \gamma \left[\sum_{i=1}^{N+1} T_i^j (\tau_{ni}^{jk} c_i^j)^{-\theta} \right]^{-1/\theta}. \quad (5)$$

- Therefore, we have

$$P_n^{j'} = \gamma \left[\sum_{i=1}^{N+1} T_i^{j'} (\tau_{ni}^{jk'} c_i^{j'})^{-\theta} \right]^{-1/\theta};$$

$$\hat{P}_n^j = \left[\sum_{i=1}^{N+1} \pi_{ni}^j \hat{T}_i^j (\hat{\tau}_{ni}^{jk} \hat{c}_i^j)^{-\theta} \right]^{-1/\theta}. \quad (20)$$

- The terms in the braskets is exactly *denominator* we calculated above.